GATE PYQs

GATE 2024

```python
def f(x, y):
    for i in range(y):
        x = x + x + y
    return x
```

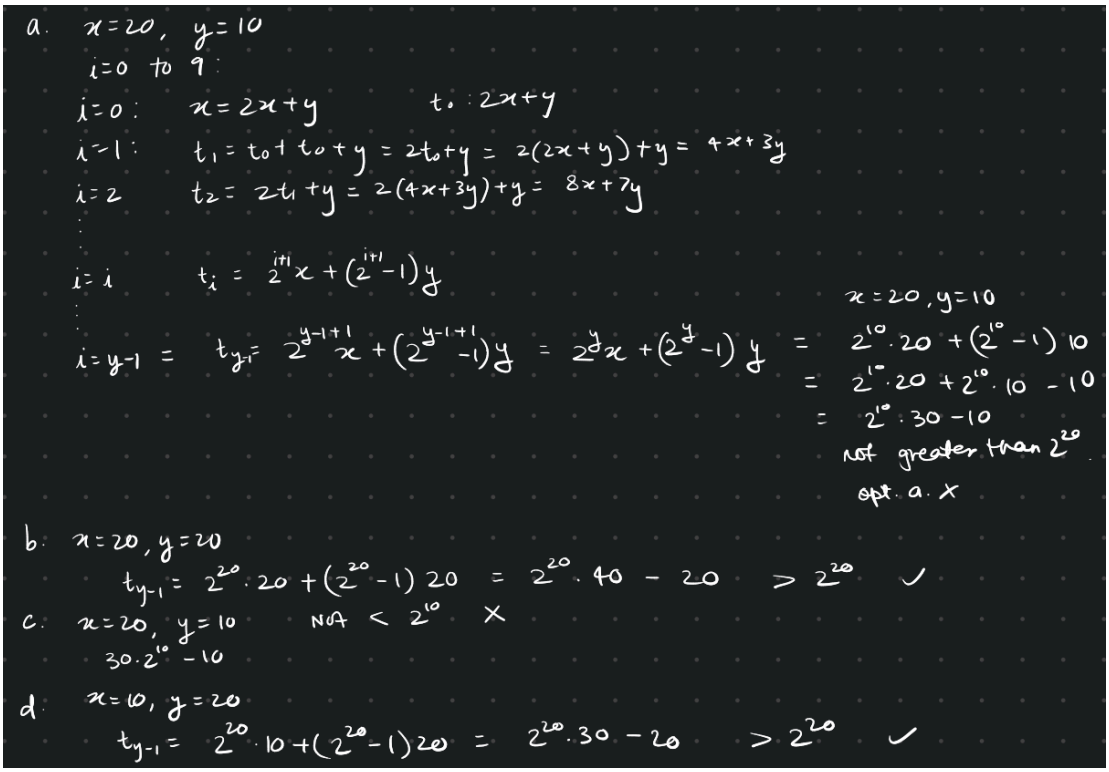
Which of the following statements is true?

a. If the inputs are x = 20 and y = 10, the return value is greater than 2^{20}.

b. If the inputs are x = 20 and y = 20, the return value is greater than 2^{20}.

c. If the inputs are x = 20 and y = 10, the return value is less than 2^{10}.

d. If the inputs are x = 10 and y = 20, the return value is greater than 2^{20}.



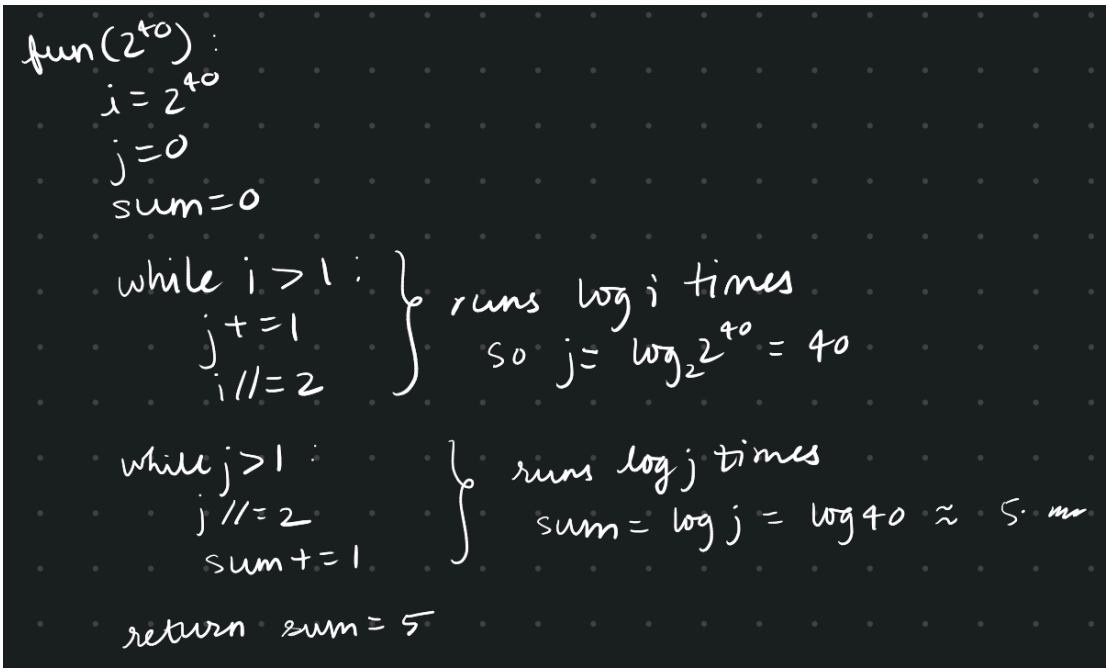
GATE 2018

```python
def fun(n):
    i = n
    j = 0
    sum = 0
    while i > 1:
        j += 1
        i //= 2
    while j > 1:
        j //= 2
        sum += 1
```

```
    return sum
```

What is the return value of the function fun when it is called with the argument 2^{40}?

ans = 5



GATE 2017

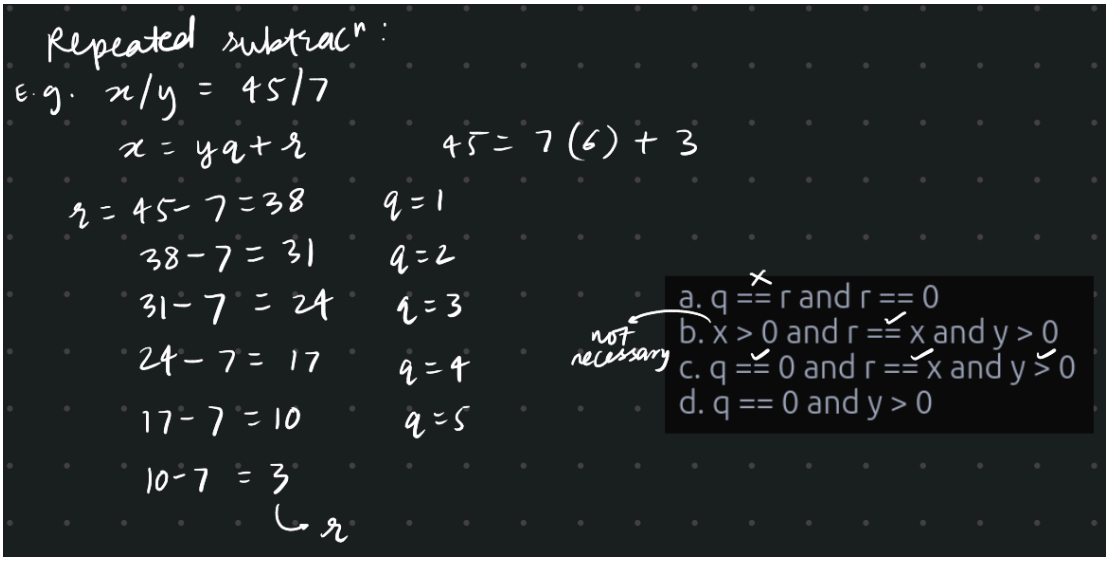
```
while r >= y:
    r = r - y
    q = q + 1
```

The code above is meant to divide two positive integers r and y using repeated subtraction.

Which of the following conditions on the variables x, y, q, and r before the code is executed will ensure that the loop terminates in a state satisfying x = yq + r?

a. q == r and r == 0
b. x > 0 and r == x and y > 0
c. q == 0 and r == x and y > 0
d. q == 0 and y > 0

opt. c

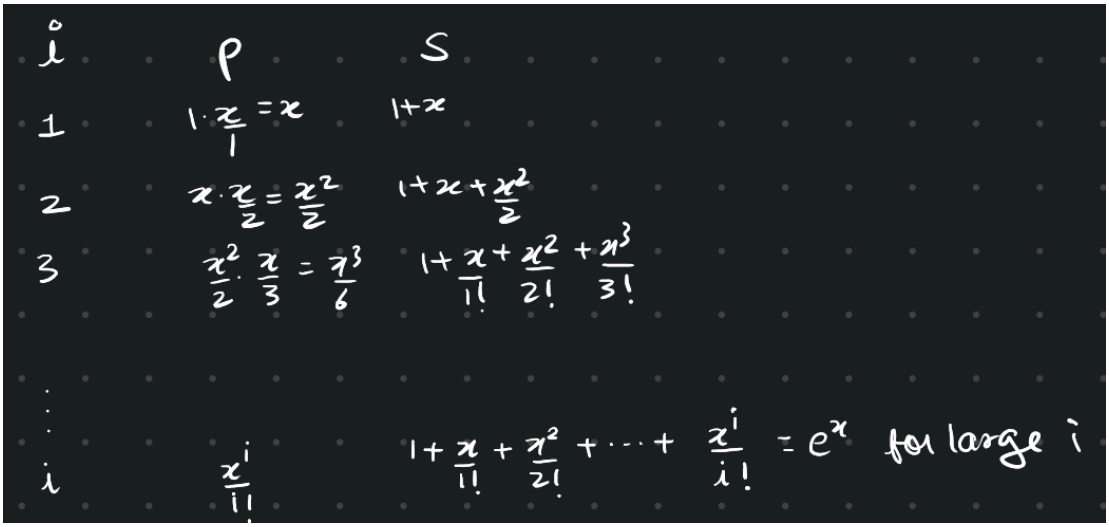


GATE 2003

```
def find_func(x, y):
    p = 1
    s = 1
    for i in range(1, y):
        p *= x/i
        s += p
    return s
```

For large values of y, the return value of the function find_func best approximates:
a. x^y b. e^x c. ln(1+x) d. x^x

ans = e^x



GATE 2021

```
def inc():
    global i
    i += 1
    return i
```

```
count = 0
i = 0

for j in range(-3, 4):
    if j >= 0 and inc():
        count += j
count += i
```

Which of the following is correct?

a. The program will throw an error.

b. Output of the program is 13.

c. Output of the program is 10.

d. Output of the program is 8.

GATE

```
def exp(X, Y):
    res = 1
    a = X
    b = Y

    while b != 0:
        if b % 2 == 0:
            a = a * a
            b = b // 2
        else:
            res = res * a
            b = b - 1

    return res
```

The function exp(X, Y) computes X^Y for positive integers X and Y. Which of the following conditions is true before every iteration of the while loop?

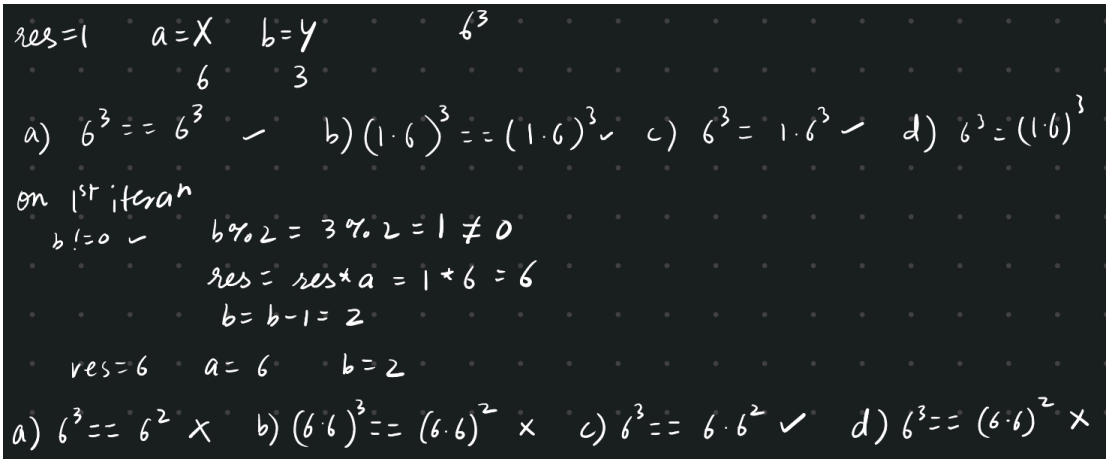
a. X^Y = a^b

b. (res * a)^Y = (res * X)^b

c. X^Y = res * a^b

d. X^Y = (res * a)^b

opt. c



GATE

```
e = 'butter'

def f(a):
    print(print(a) or e)

f('bitter')
```

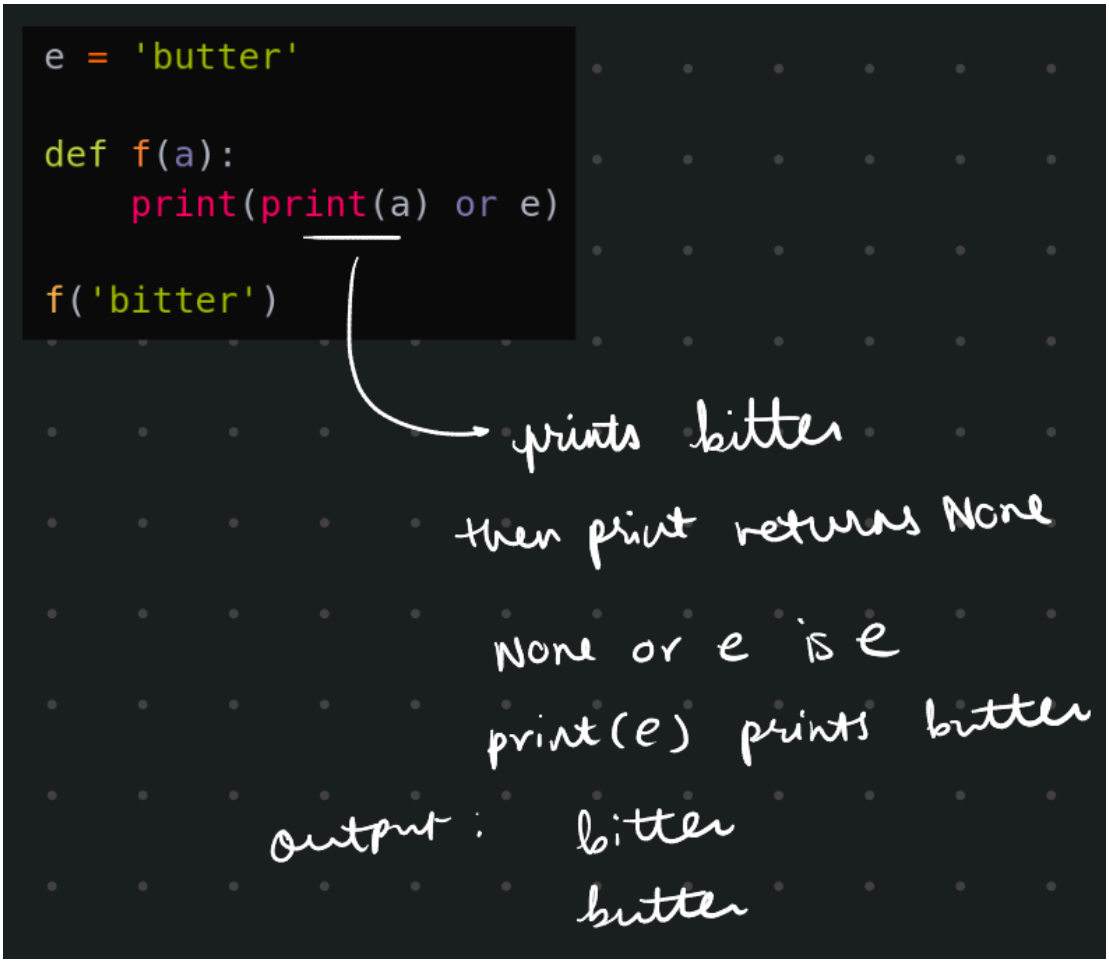
What is the output of the code?

a. error

b. butter

c. bitter

d. butter bitter



ISRO 2020

```
name = 'satellites'

l = len(name)
s = len(set(name))

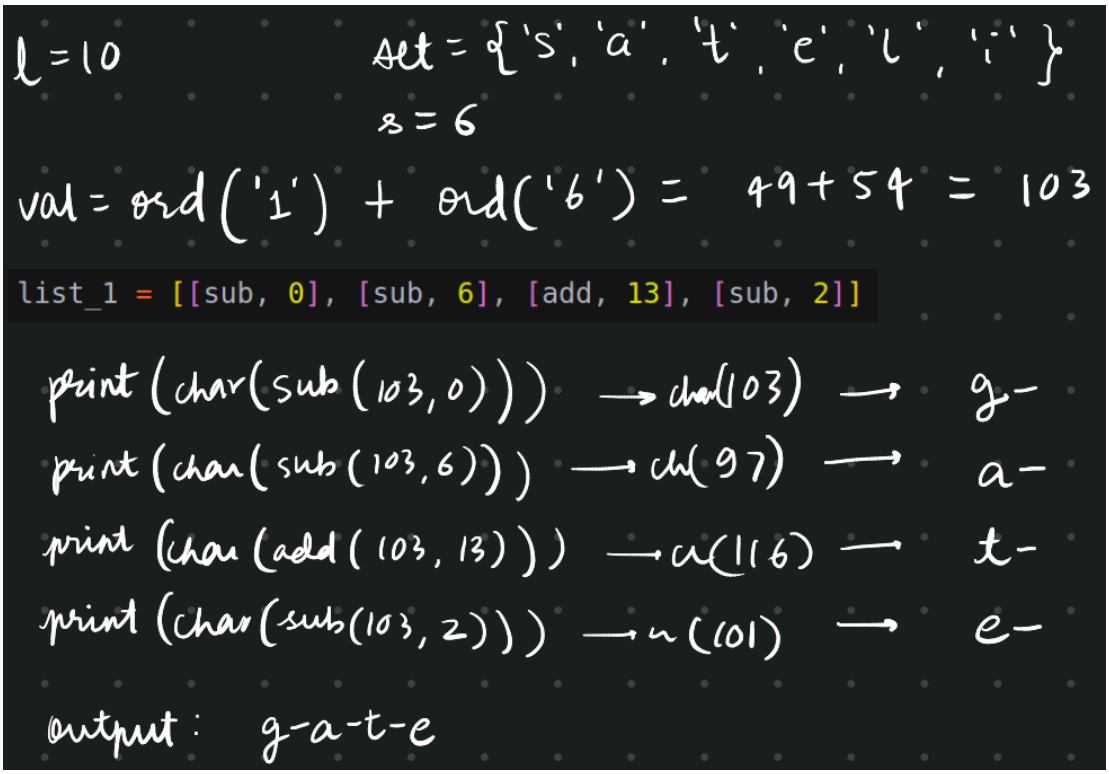
val = ord(str(l)[0]) + ord(str(s))

def add(a, b):
    return a + b
```

```
def sub(a, b):
    return a - b

list_1 = [[sub, 0], [sub, 6], [add, 13], [sub, 2]]

for i in range(len(list_1)):
    print(chr(list_1[i][0](val, list_1[i][1])), end='-')
```



GATE 2014

```
def myX(E: list, size):
    Y = 0
    for i in range(size):
        Y = Y + E[i]

    for i in range(size):
        for j in range(i, size):
            Z = 0
            for k in range(i, j+1):
                Z = Z + E[k]
            if Z > Y:
                Y = Z
    return Y
```

The value returned by myX is

a. the maximum possible sum of elements in any sub-list of E

b. the maximum element in any sub-list of E

c. sum of the maximum elements in all sub-lists of E

d. sum of all elements in E

opt. a

[1, 2, 3, 4], 4

y = 0 for i in range(4):
        y += E[i]
        y = 1 + 2 + 3 + 4 = 10

y = 10

i = 0 : j = 0 1 2 3

	k = 0	k = 0, 1	k = 0, 1, 2	k = 0, 1, 2, 3
	z = 1	z = 3	z = 6	z = 10

i = 1 : j = 1 j = 2 j = 3

	k = 1	k = 1, 2	k = 1, 2, 3
	z = 2	z = 2 + 3	z = 2 + 3 + 4

So essentially
z takes values of all sums of subarrays one-by-one and
y takes the max of them.
opt. d. turns out incorrect for -ve #s.